Energy in Quantum-Dot Cellular Automata

Point of contact: Dr. Austin Luchsinger

Computer Science, University of Texas Rio Grande Valley austin.luchsinger@utrgv.edu

Background

Quantum dots confine electrons in tiny regions; their discovery and synthesis earned the 2023 Nobel Prize in Chemistry [1]. *Quantum-dot cellular automata* (QDCA) process classical information with interacting dot arrangements. In a common cell, two electrons favor opposite corners of a square; its two diagonals encode 0 and 1. Bringing electrons closer costs electrostatic energy [2].

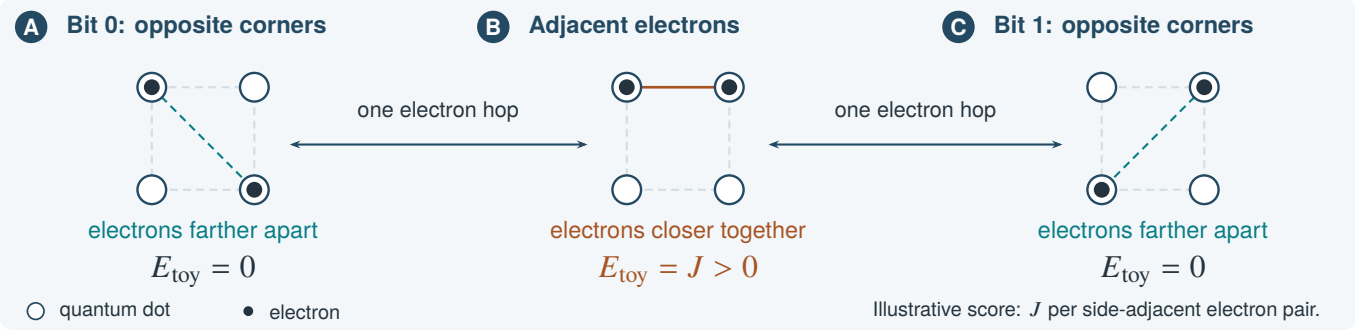


Figure 1. Three occupations of one four-dot, two-electron cell. Under single-electron edge hops, changing the diagonal passes through an adjacent pair. This toy score omits tunneling physics and external fields.

QDCA designs implement Rule 110 and Conway's Game of Life [3,4], and physical energy models already exist [2]. The challenge here is a *simple, discrete* energy model and a precise connection between favorable moves and correct computation.

The problem

1. **Define a simplified energy model.** Assign $E(C)$ to charge configurations: for example, penalize nearby electron pairs, possibly with distance-dependent weights. Specify geometry, electron counts, local moves, fixed inputs, and external controls. Test a single cell and a short wire or gate. Treat the model as a classical approximation and identify omitted physical effects.
2. **Simulate a cellular automaton using favorable transformations.** Can Rule 110 or Game of Life evolve under your energy rules? One first target permits only moves with $E(C') \leq E(C)$. Define how states encode a generation and how the next is produced. Correct low-energy outputs are not enough: do the allowed paths reach them? Make clocking, changing inputs, or energy-increasing moves and their energy contribution explicit.

A starting point

Explain the switching obstruction in Figure 1 under its stated hopping rule. Then check one Rule 110 update on all eight input triples before attempting multiple cells and generations. Aim for a verified gadget, a small energy-respecting simulation, or an obstruction to a proposed rule. AI or exhaustive search can help; distinguish finite tests from a human-readable general argument.

References

[1] Royal Swedish Academy of Sciences. The Nobel Prize in Chemistry 2023: Press release. October 4, 2023.

[2] C. S. Lent, P. D. Tougaw, W. Porod, and G. H. Bernstein. Quantum cellular automata. *Nanotechnology* 4:49–57, 1993.

[3] D. Tougaw and J. D. Will. Designing a Turing-complete cellular automata system using quantum-dot cellular automata. *J. Comput. Electron.* 19:1337–1343, 2020.

[4] O. Liolis, G. Ch. Sirakoulis, and A. Adamatzky. Conway's Game of Life in Quantum-dot Cellular Automata. *Microelectronics Journal* 109:104972, 2021.